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B.Tech. Degree IV Semester Supplementary Examination April 2021

ME 15-1402 METROLOGY AND INSTRUMENTATION (2015 Scheme)

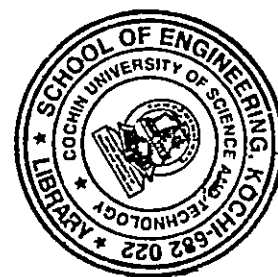
Time : 3 Hours

Maximum Marks : 60

PART A (Answer ALL questions)

(10 × 2 = 20)

- I. (a) What are Systematic errors? Can we eliminate these errors?
- (b) Discuss the effect of supports on deformation of measuring bars.
- (c) What do you mean by 'hole basis system'? Briefly explain.
- (d) Explain the importance of an optical flat.
- (e) State the essential properties of light waves for interference.
- (f) What do you mean by lay? Give symbols of different types of lay on surface.
- (g) What is static calibration? Explain.
- (h) Distinguish between active and passive transducer.
- (i) What is a sound level meter? What are its basic parts?
- (j) What are the various pollutants in air? Mention different methods used to find out air quality.



PART B

(4 × 10 = 40)

- II. (a) For a measuring instrument, discuss: (5)
(i) Sensitivity (ii) Magnification (iii) Calibration (iv) Precision (v) Traceability
- (b) Which agency maintains standards internationally? Explain why material standards are replaced by non-material standards. Use examples to substantiate your points. (5)

OR

- III. (a) Compare Clearance fit and Interference fit. Give examples. (5)
- (b) Explain the significance of Tolerance and Interchangeability in mass production. (5)
- IV. (a) Write a short note on spirit level. (5)
- (b) In the measurement of surface roughness, heights of 20 successive peaks and valleys measured from datum are as follows: (5)
45, 25, 40, 25, 35, 16, 40, 22, 25, 34, 25, 40, 20, 36, 28, 18, 20, 25, 30, 38.
If these measurements were made over a length of 20 mm, determine the C.L.A and R.M.S values of the surface.

OR

- V. Explain the working of following instruments with neat sketch (10)
(i) NPL flatness type interferometer (ii) Laser beam interferometer
- VI. What are the different methods of correction for interfering and modifying inputs? Explain with examples. (10)

OR

- VII. (a) Explain the generalized measurement system with block diagrams. Give an example. (6)
- (b) Describe the working of potentiometer transducer as zero order instrument. (4)
- VIII. (a) Explain the working of Scintillation Counter with neat sketches. (5)
- (b) Discuss five laws of thermocouple. (5)

OR

- IX. (a) Explain any temperature measurement process without involving physical contact, in detail. (5)
- (b) What do you mean by temperature compensation in strain gauge? Why is it required? (5)